

Freight rate prediction

Machine Learning Engineer assessment: data, validation design, model and results

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1 Summary

The task is to predict the posted rate of 12,000 truckload shipments dated November and December 2025, given 48,000 labelled loads dated January to October 2025. Every load to predict is dated after the last labelled load, so I treated this as a two-month-ahead forecast and validated it as one: model selection used rolling-origin backtests, each fold training on all earlier months and testing on the next two, never a random split.

The final model is a hybrid. A small linear component owns the time dimension (log market index and its square, a linear trend, weekday and quarter-end terms) and a LightGBM model fits the remaining load-level pricing from distance, equipment, weight and coordinates. On the backtest fold that mirrors the real task (fit on January to August, test on September and October) it reaches a mean absolute error of \$39 and a mean absolute percentage error of 1.7% on loads with clean labels. Averaged over the three folds the MAE is \$38, against \$51 for the best plain LightGBM configuration, \$47 for a log-linear regression and \$167 for a rate-per-mile baseline.

Four data problems shaped the work: sign-flipped and missing weights, missing market index values, 1.4% of labels multiplied or divided by a factor between two and six, and a `quote_signal` column that is a near copy of the target in some months, its mirror image in others, and pure noise in the prediction window. Eight of the 72 validation cities never appear in training, which is why the model prices geography through coordinates rather than city names.

2 Data

2.1 What the files contain

	Training (<code>train_test.csv</code>)	To predict (<code>validation.csv</code>)
Loads	48,000	12,000
Dates	2025-01-01 to 2025-10-31, 304 days	2025-11-01 to 2025-12-31, 61 days
Loads per day	about 158	about 197
Cities	64	72, eight of them unseen in training
Lanes (pickup, delivery)	4,014	4,214, of which 736 unseen
Equipment	Dry Van 57%, Reefer 25%, Flatbed 18%	same mix
Distance	70 to 3,440 miles, median 953	same range

Each load carries pickup and delivery city with coordinates, road distance, equipment type, weight, date, `market_index` and `quote_signal`. The coordinates are consistent (each city has exactly one pair) and the distance is about 1.18 times the great-circle distance with a floor at 70 miles, so neither needed cleaning. The December chart template carries only city names, distance, equipment, weight and date.

2.2 Quality issues and how I handled them

Issue	Extent	Treatment
Negative weights	292 training loads, 145 validation loads	Sign flip: the magnitudes follow the same distribution as positive weights and the rates match heavy loads. Absolute value taken.
Missing weights	300 and 165 loads	Left as missing; LightGBM handles it. The affected loads are otherwise unremarkable.
Missing market_index	374 and 249 loads	Replaced by the daily mean of the index, see below.
Corrupted labels	677 training loads (1.4%)	Rate multiplied by 2.2 to 5.3 (340 loads) or divided by 2.2 to 5.9 (337 loads), spread evenly over months and equipment. Excluded from training, reported separately in evaluation.
quote_signal	whole column	Dropped. Its relationship with the target changes sign by month and vanishes in the prediction window (section 2.4).
Unseen cities	1,447 validation loads (12%) touch one	Geography enters the model only through coordinates; a held-out-city check quantifies the cost (section 4.4).

The corrupted labels are easy to isolate. Figure 1 shows the posted rate relative to a deliberately simple log-linear reference fit (distance, equipment, weight, market index, trend). Clean loads sit within about $\pm 25\%$ of the reference; the corrupted ones sit at 0.17 to 0.45 or 2.2 to 5.3 times it, with nothing in between, so any cut-off in the gaps gives the same set. The hidden validation labels presumably carry the same corruption. Nothing can be done about that at prediction time, so the sensible target is the clean rate, and I report errors both with and without the corrupted rows.

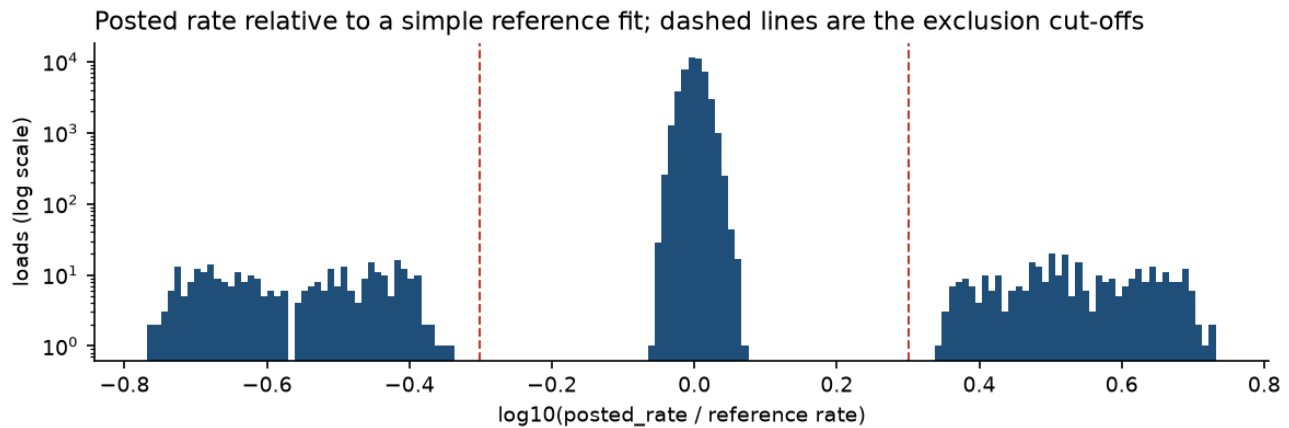


Figure 1: Posted rate divided by a simple reference fit, log scale. The two dashed lines are the exclusion cut-offs; 677 loads fall outside them.

2.3 Structure worth modelling

Load-level pricing. Rates are close to a rate per mile times distance, with a volume discount: log rate rises with log distance with a slope of 0.87, so the rate per mile falls from about \$2.97 under 100 miles to \$1.89 above 2,500 miles. Reefer loads pay 13% more per mile than dry vans and flatbeds 8% more. Heavier loads pay more, about 7% from the lightest to the heaviest bucket, identically for all equipment types. Cities carry stable premiums of up to $\pm 3.5\%$ (the same in the first and second half of the year, correlation 0.98), and those premiums track latitude: southern origins and destinations pay more (Figure 5). That is what makes coordinates a safe way to price the eight cities the training data has never seen.

Time. Figure 2 shows the daily mean rate per mile and the daily mean market index. Three things stand out. The market index is a market-wide daily series with a strong weekly cycle (Monday

1.00 to Thursday 1.18 on average) on a slowly shifting level; the per-load scatter around the daily mean has a standard deviation of 0.025 and no correlation with the load's rate (0.03), so the daily mean is the useful quantity and it fills the missing values for free. Rates follow the index (daily correlation 0.65) but also drift upward through the year and show a premium in the last two to three weeks of every quarter (Figure 3), peaking at about 5% in the final three days. The quarter-end effect matters for December. Holidays showed no effect beyond the weekday pattern, so I did not add holiday features.

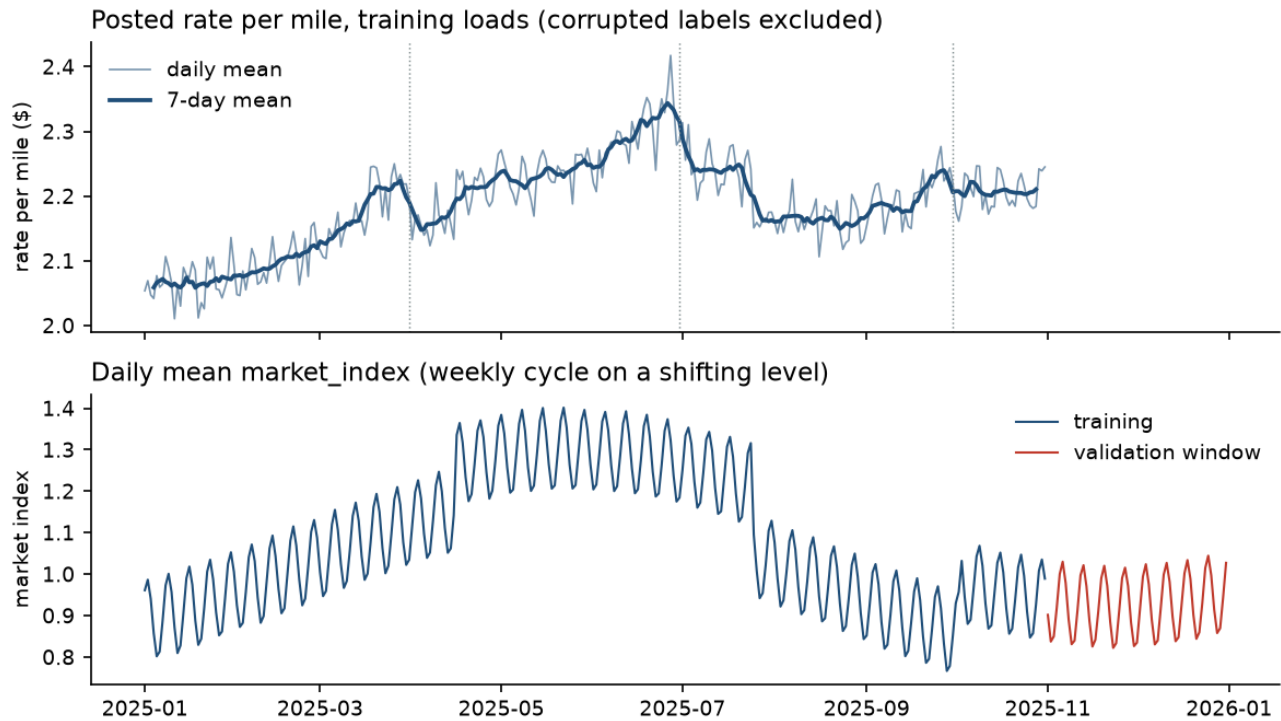


Figure 2: Daily mean rate per mile on clean training loads (top) and daily mean market index for the training and prediction windows (bottom). Dotted lines mark quarter ends.

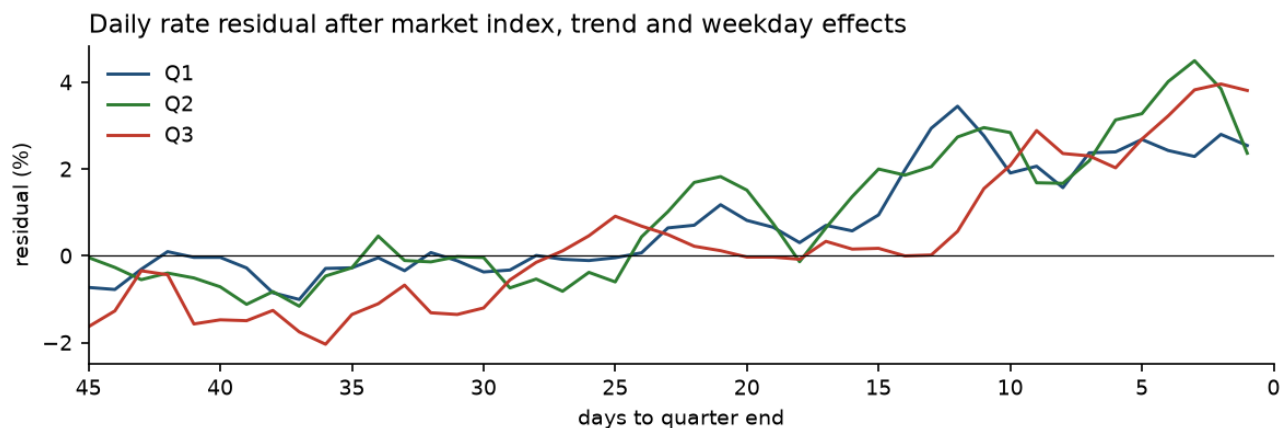


Figure 3: Daily residual of the rate per mile after market index, trend and weekday effects, against days to quarter end, for the three complete quarters.

2.4 The quote_signal column

quote_signal looks like a rate-per-mile quote (mean 2.06, close to the observed 2.22) and correlates with the target at 0.05 overall, which would normally make it forgettable. Split by month it is anything but (Figure 4, left). In January, February, March, June and September its correlation with the rate per mile is 0.99: the two agree to within a standard deviation of 0.025, so it is essentially the label. In April, May, July and October the correlation is -0.99 : the column equals 4.15 minus the rate per mile, a mirror image. In August it is uncorrelated noise. The regime flips at month boundaries with no pattern I could find.

Which regime the prediction window is in decides whether the column is useful at all, and the validation file answers that without needing labels. In every informative month the column separates by equipment, because reefer and flatbed loads price differently. In August, November and December the mean quote_signal is 2.05 for all three equipment types and its spread (0.22) is the smallest of any month (Figure 4, right). The prediction window is a noise regime. A model that had learned to use the column would either apply October's mirrored relationship to noise or, at best, ignore it; it cannot help. I dropped it. Section 3.4 shows what the split choice would have said about this.

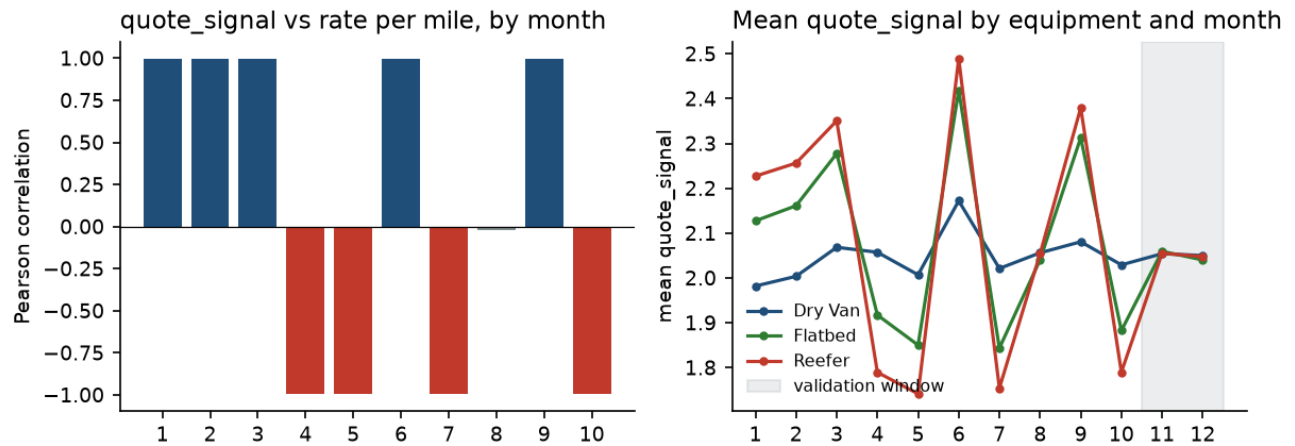


Figure 4: Left: monthly Pearson correlation between quote_signal and the rate per mile. Right: mean quote_signal by equipment and month; in informative months the equipment types separate, in August and the prediction window they do not.

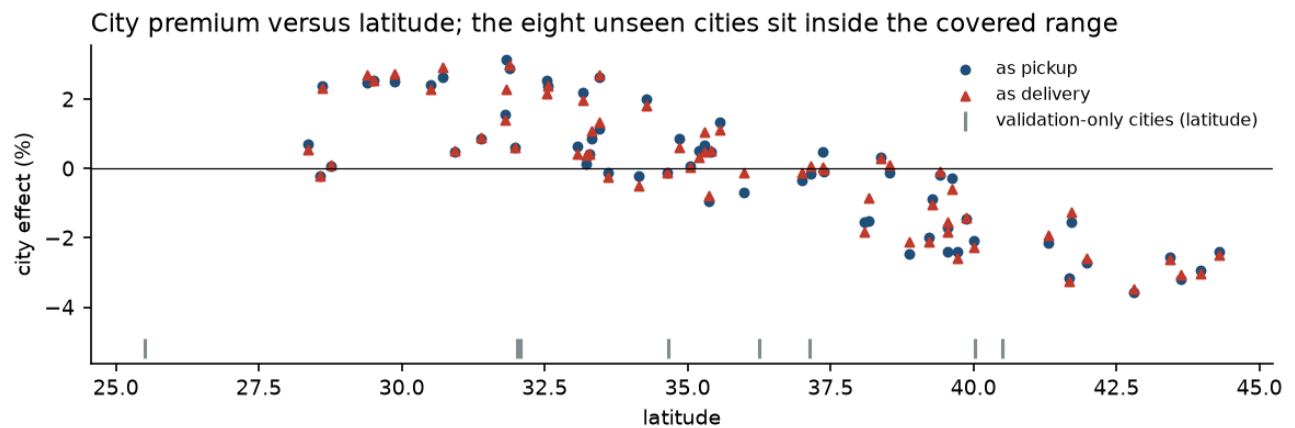


Figure 5: Mean residual of the reference fit by city (as pickup and as delivery) against latitude. The ticks at the bottom mark the latitudes of the eight cities that only appear in the validation set.

3 Validation design

3.1 Why a time-based split

The prediction window starts the day after the training window ends and runs for two months. A random split would let the model see loads from the same week on both sides, which measures interpolation, not the forecast the task asks for. Three things in this data only show up under a time-based split: a tree's inability to extrapolate the trend, the confounding between market index and season (both rise from January to May), and the `quote_signal` regime changes. All three would have been invisible under random cross-validation and all three mattered for the final model.

3.2 Folds

I used three rolling-origin folds, each training on every load before the test window and testing on the next two months, with corrupted labels removed from the training side:

Fold	Fit on	Test on	Test loads
1	January to April	May and June	9,689
2	January to June	July and August	9,719
3	January to August	September and October	9,523

Fold 3 has the same shape as the real task. Folds 1 and 2 guard against tuning to one particular pair of months; the market index steps down sharply in mid July, for instance, which only fold 2 tests. Model choices were made on the mean over folds, and the final model was refitted on all clean training loads with the settings fixed by the backtests.

3.3 Metrics

Spotter's metric is not disclosed, so I report MAE, RMSE, MAPE and R^2 in dollars of posted rate. All are computed on test loads with clean labels (the quantity the model is meant to predict) and MAE and MAPE additionally on all test loads including the corrupted ones, since the hidden validation labels presumably include them. The corrupted rows dominate RMSE on the full set and set a floor no model can beat, which is why the clean-label numbers are the ones used for selection.

3.4 What a random split would have said

The check in `backtest.quote_signal_check` fits the same LightGBM model with and without `quote_signal`, once under random 5-fold cross-validation and once on the time-based folds, reporting the MAE per test month:

Evaluation	MAE without <code>quote_signal</code>	MAE with <code>quote_signal</code>	Change
random 5-fold, all months	38.0	35.0	-8%
time-based, test May	43.2	37.6	-13%
time-based, test Jun	74.7	84.5	+13%
time-based, test Jul	45.9	40.1	-13%
time-based, test Aug	61.9	68.7	+11%
time-based, test Sep	43.1	40.3	-6%
time-based, test Oct	60.1	55.8	-7%

Under a random split the column looks useful (8% lower error). Under the time-based folds it helps in some months and hurts in others by up to 13%, depending on whether the test month happens to share the regime of the months before it. Since the prediction window is a noise regime, the right expectation for November and December is no gain at all, and the risk is on the downside.

4 Model

4.1 Candidates

All models predict log posted rate and are evaluated in dollars. The features common to all learned models are log distance, equipment, weight, the four coordinates, day of week, days to quarter end and day of quarter. Month and day of year are deliberately excluded: November and December never occur in training, and a tree would map them onto whichever leaf happens to be adjacent. Day of week and position within the quarter take the same values in the fourth quarter as in the first three.

- *Median rate per mile by equipment, times distance.* The naive baseline.
- *Log-linear regression.* The time design below plus linear terms in log distance (and its square), equipment, weight and coordinates.
- *LightGBM, load and calendar features.* No market or trend information.
- *LightGBM plus daily market index.*
- *LightGBM plus daily market index plus a day index.* The day index lets the tree hold the level of the last training weeks for later dates.
- *Hybrid.* Described next.

4.2 The hybrid

A gradient-boosted tree has two problems with the time dimension here. It cannot extrapolate: for any date past the training window it returns the level of the last leaf. And within the training window the market index and the season are confounded, so it may attribute the spring rise to the index and then, seeing similar index values in September, predict February-like rates. In the backtests the plain tree with the market index was worse than the tree without it on the last fold for exactly this reason.

The hybrid separates the two roles. A linear model owns the time component:

$$\log(\text{rate}) = a + \beta_1 \log m + \beta_2 (\log m)^2 + \gamma t + \delta_{\text{weekday}} + \theta_{\text{quarter-end bucket}} + f(\text{load})$$

where m is the daily mean market index, t is time in years and the quarter-end buckets are 0 to 3, 4 to 7, 8 to 14, 15 to 21 and 22 to 30 days before quarter end. f is a LightGBM model on log distance, equipment, weight, coordinates, weekday and days to quarter end. The two parts are fitted by backfitting: the linear part on log rate minus the current tree prediction, then the tree on log rate minus the linear part, repeated twice. A third pass did not improve the backtests and a single pass was clearly worse. The trend term exists only in the linear part, so the level extrapolates as a straight line in log space rather than freezing, and the market index enters with a fitted elasticity instead of a learned lookup.

The fitted time component on all training data is readable:

Term	Coefficient (log scale)	Meaning
log market index	0.141	elasticity of about 0.14 at index 1.0
(log market index) ²	0.115	rates react more strongly when the index is high (elasticity 0.20 at 1.3, 0.10 at 0.85)
trend	0.064 per year	about 0.5% per month
quarter end, last 3 days	0.052	+5.3%
quarter end, 4 to 7 days	0.033	+3.3%
quarter end, 8 to 14 days	0.016	+1.6%
quarter end, 15 to 21 days	0.021	+2.1%
quarter end, 22 to 30 days	0.003	negligible
weekday terms	within ± 0.01	the weekly cycle is carried by the index

Adding the squared index term improved every fold. Lagged versions of the index (7, 30 and 60 day means), a start-of-quarter dip, daily rather than bucketed quarter-end dummies and

equipment-specific time terms were all tested and rejected: none improved the mean over folds and most made at least one fold worse. LightGBM settings (31 leaves, minimum 40 loads per leaf, learning rate 0.03, 1,500 rounds, 80% feature and row subsampling) were chosen from a small grid; the alternatives moved the mean MAE by about a dollar either way.

4.3 Results

Mean absolute error in dollars on test loads with clean labels, per fold and averaged:

Model	Fold 1 (May-Jun)	Fold 2 (Jul-Aug)	Fold 3 (Sep-Oct)	Mean
Median rate per mile by equipment	153.7	172.4	175.1	167.1
Log-linear regression	51.1	43.6	47.7	47.5
LightGBM, load and calendar features	135.7	72.4	70.6	92.9
LightGBM, plus market index	52.4	62.8	86.3	67.2
LightGBM, plus market index and day index	54.5	51.1	48.0	51.2
Hybrid (final)	38.6	35.1	39.1	37.6

Other metrics, averaged over the three folds. The last two columns include the corrupted test labels:

Model	RMSE	MAPE	R ²	MAE, all rows	MAPE, all rows
Median rate per mile by equipment	234.0	8.33%	0.9702	219.3	10.53%
Log-linear regression	68.2	2.05%	0.9975	101.4	4.40%
LightGBM, load and calendar features	122.1	3.84%	0.9917	146.2	6.15%
LightGBM, plus market index	93.6	2.83%	0.9952	120.8	5.12%
LightGBM, plus market index and day index	73.9	2.19%	0.9971	105.1	4.54%
Hybrid (final)	54.8	1.63%	0.9984	91.6	3.99%

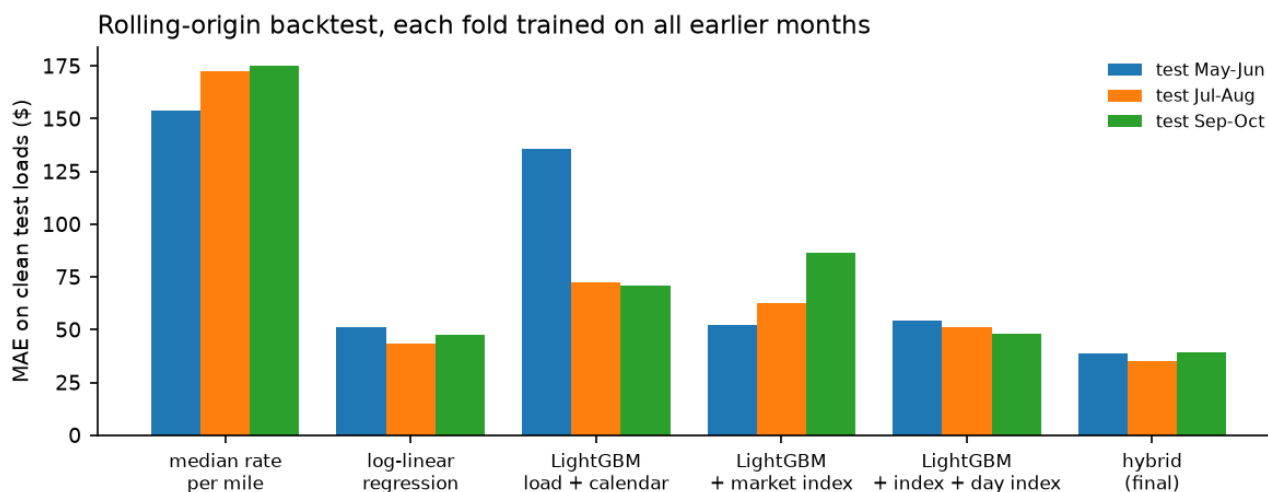


Figure 6: MAE on clean test loads per fold and model.

The ordering is the same on every fold. Two things are worth noting. The log-linear regression beats every plain LightGBM configuration, which says the hard part of this problem is the time

structure, not the load-level non-linearity. And the pure tree with the market index but no day index is the worst learned model on the last fold, the confounding failure described above.

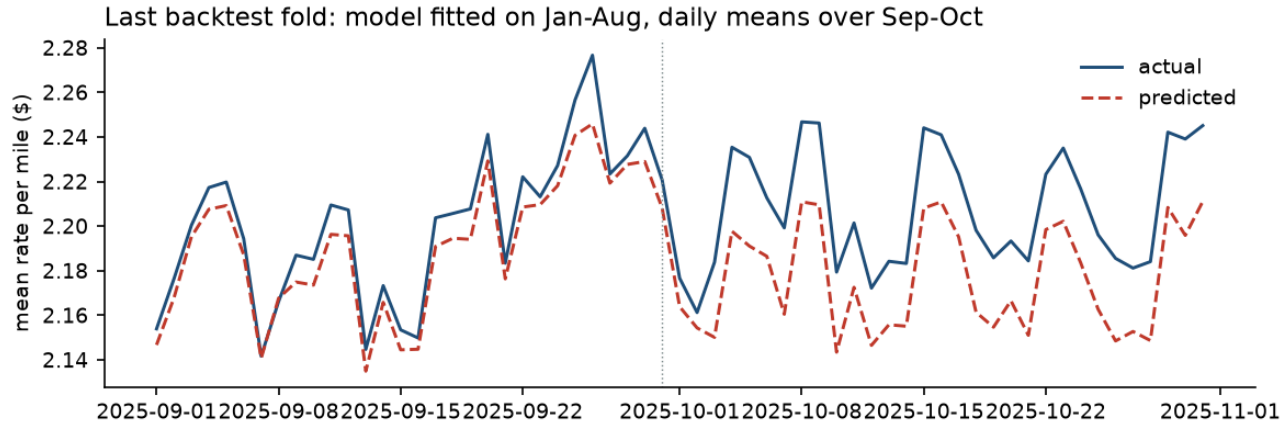


Figure 7: Last fold: daily mean rate per mile of the actual September and October loads against the hybrid’s predictions, fitted on January to August only. The dotted line is the quarter end.

Figure 7 shows the hybrid tracking the daily level through the fold-3 test window, including the September quarter-end rise and the drop on October 1, neither of which is in its training window at those dates. The residual bias over the window is small but not zero: the model runs about 0.4% low in September and 1.4% low in October, so the extrapolated level errs slightly on the conservative side as the horizon grows.

4.4 Unseen cities

Twelve percent of the validation loads touch one of eight cities absent from training. To measure the cost, I refitted the hybrid on fold 3 with six randomly chosen training cities (Greensboro, Milwaukee, Salt Lake City, Houston, New York, Amarillo) removed and compared test loads that touch them against the rest:

Test loads	n	MAE	RMSE	MAPE	R ²
loads touching held-out cities	1,672	46.0	64.8	2.14%	0.9974
other loads	7,707	40.1	57.8	1.77%	0.9983

Loads involving an unseen city cost about 0.4 points of MAPE, which is the price of pricing geography from coordinates alone rather than a city identifier. The eight real unseen cities sit inside the latitude and longitude range of the training cities (Figure 5), so this estimate should transfer.

5 December series

The chart template fixes the lane (Lexington to Fort Wayne, 360 miles, dry van, 32,000 lb) and varies only the date. The template carries no coordinates or market index, so the coordinates come from the city lookup and the market index for each December day is the mean market_index of the roughly 200 validation loads dated that day, exactly as it is for every other prediction. The prediction therefore shows the time effects the model learned: the weekly cycle carried by the index (Thursday peaks), the slow drift, and the quarter-end premium in the last week, ending at \$869 on December 31 against \$828 on December 1.

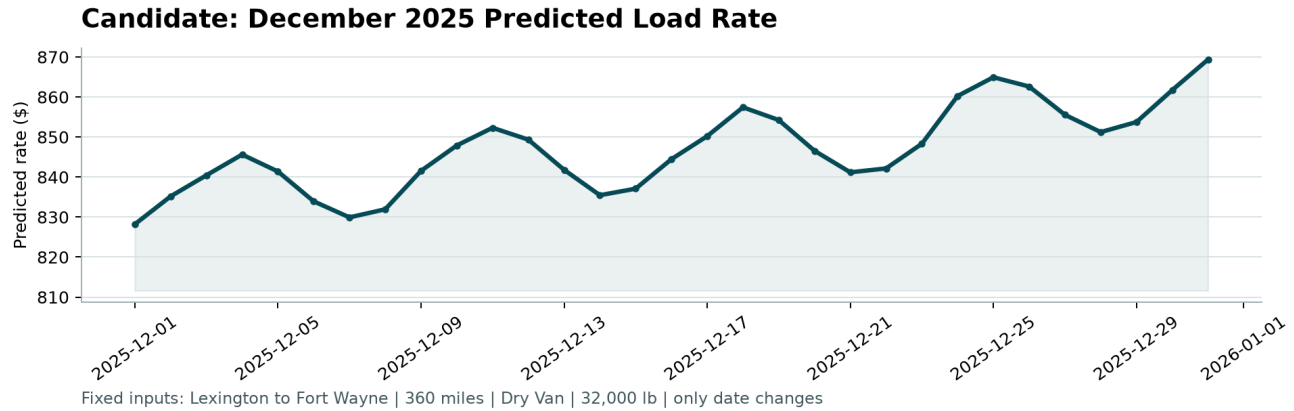


Figure 8: Chart produced by `score.py` from `outputs/december_chart_inputs.csv`.

The level is consistent with the lane’s history: the same lane and equipment priced between \$778 and \$865 in September and October, and the model’s in-sample predictions for those loads are within 1 to 3%.

6 Reproducing the results

The repository contains the data, the code, and the outputs. From a clean checkout:

```
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python -m pytest tests -q
python -m freight.train --backtest
python -m freight.predict
python score.py --predictions outputs/validation_predictions.csv \
                --december-predictions outputs/december_chart_inputs.csv \
                --output-dir outputs/scorer_results
```

Training with backtests takes about a minute on a laptop; the final fit alone takes a few seconds. Results are deterministic. The unit tests cover the cleaning rules, the calendar features on December dates, the label screening on planted corruption, and that the hybrid recovers a known elasticity, trend and quarter-end effect from synthetic data.

7 Limitations

The trend is extrapolated as a straight line in log space for two months beyond the data. The three folds support that choice (a model without the trend was much worse on every fold, and lagged market indices did not replace it), but a year-end break in the drift would not be visible from ten months of history. The quarter-end premium is assumed to apply to December as it did to March, June and September; the December estimate is the average of those three. Finally, if Spotter’s metric is RMSE on all loads including the corrupted ones, predictions could be nudged up by about 1% to hedge the asymmetric corruption; I chose not to, since the clean rate is the quantity that is actually useful.